

NAME SCHOOL TEACHER 

Pre-Junior Certificate Examination, 2017

Mathematics

Paper 2
Ordinary Level

Time: 2 hours

300 marks

School stamp

Running total	
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For examiner			
Question	Mark	Question	Mark
1		11	
2		12	
3			
4			
5			
6			
7			
8			
9			
10		Total	

Grade

Instructions

There are 12 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. You may ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You will lose marks if you do not show all necessary work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:

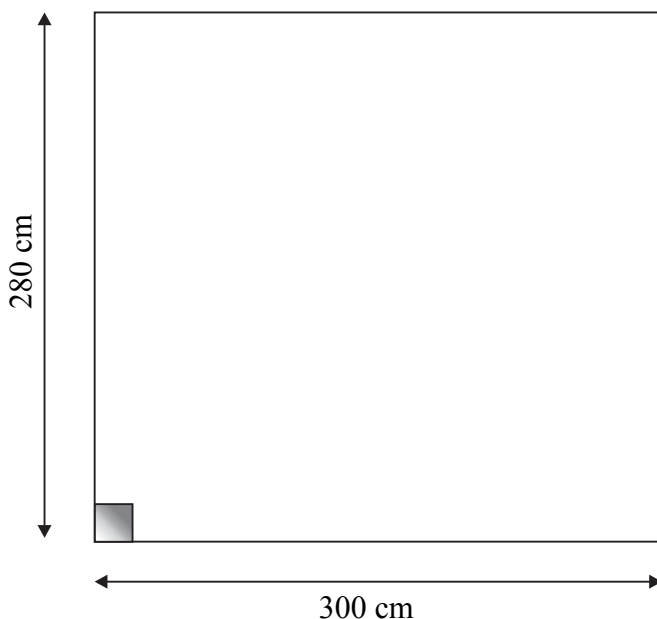
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Question 1

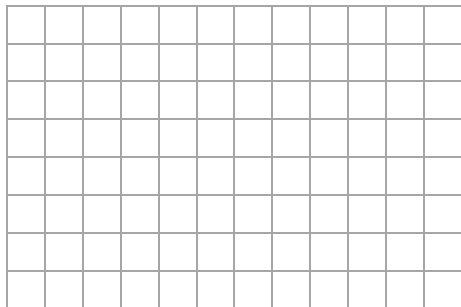
(Suggested maximum time: 10 minutes)

Mark is tiling a wall in his bathroom.
The wall is in the shape of a rectangle measuring 300 cm by 280 cm.

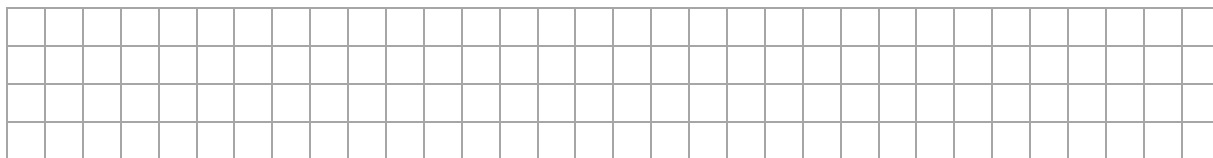
The diagram shows one of the square tiles with side length 20 cm which Mark wishes to tile the wall with.



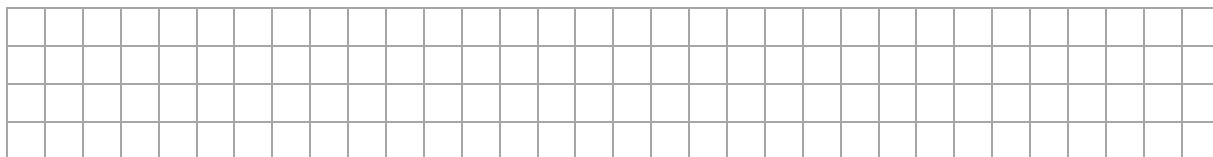
- (a) Find the **area** of the wall.



- (b) Find the **area** of one tile.



- (c) Find the **least number** of tiles Mark needs to tile the wall.

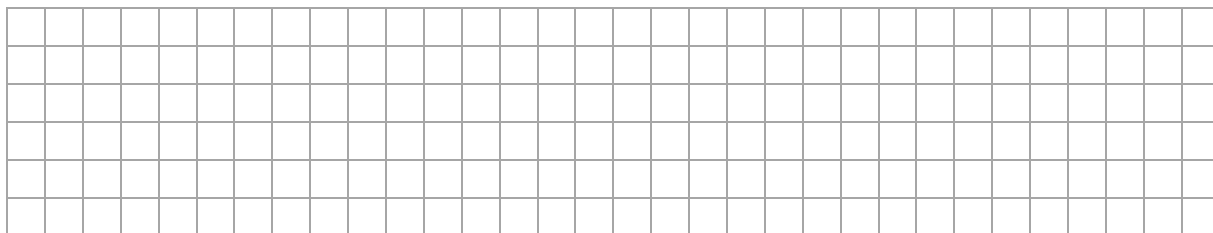


- (d) There are 25 tiles in a box. Mark says: "I will need 8 boxes to tile the wall".
Is Mark correct? Justify your answer fully.

Answer: _____

Justification: _____

- (e) Mark started to tile the wall at 9:15 a.m. He finished the job at 12:05 p.m. the same day.
How long did it take Mark to tile the wall? Give your answer in hours and minutes.



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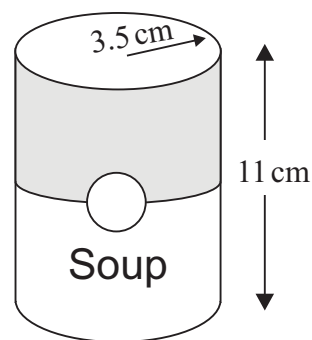
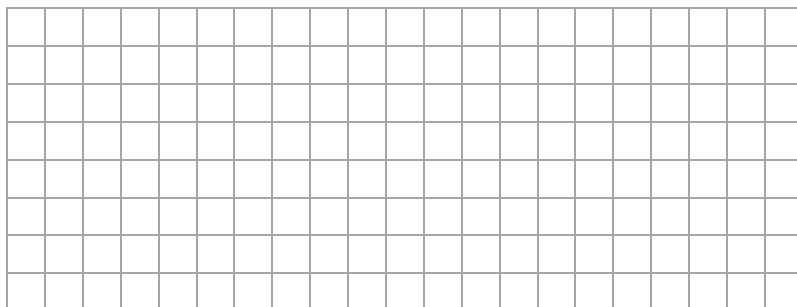
Question 2

(Suggested maximum time: 10 minutes)

- (a) A cylindrical tin of soup has a radius of 3.5 cm and a height of 11 cm as shown in the diagram.

Find the **volume** of the tin.

Give your answer correct to the nearest cm^3 .

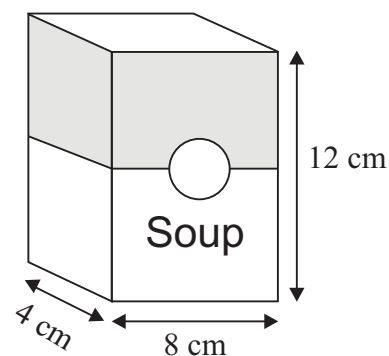
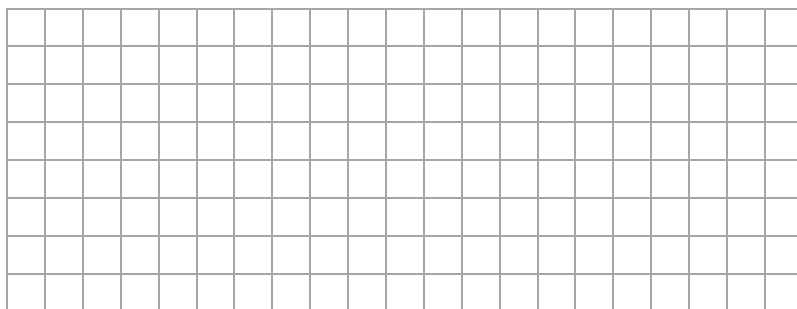


- (b) The same brand of soup can be purchased in a rectangular carton.

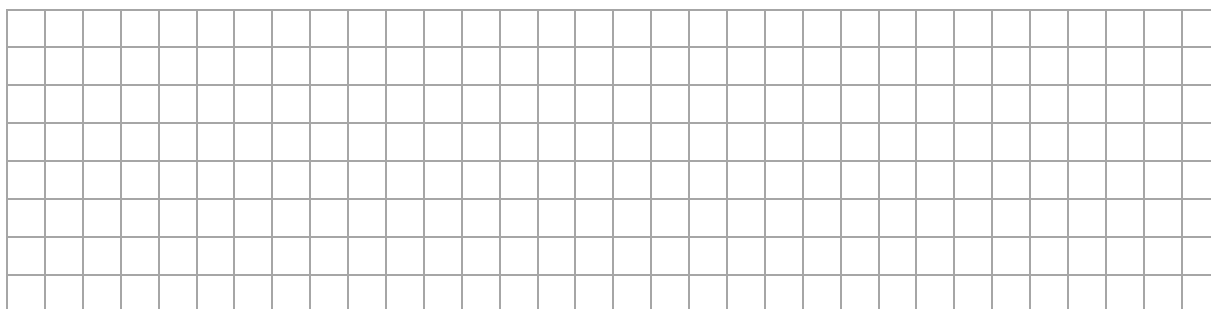
The carton has a length of 8 cm, a width of 4 cm and a height of 12 cm, as shown.

Find the **volume** of the carton.

Give your answer correct to the nearest cm^3 .



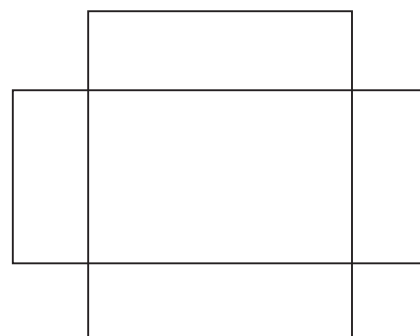
- (c) Derek wishes to buy 2 litres of the soup. The shop only sells soup in the rectangular cartons. How many cartons does he need to buy? Note: $1000 \text{ cm}^3 = 1 \text{ litre}$.



- (d) Derek says that the net of this rectangular carton is as shown. Is he correct? Give a reason for your answer.

Answer: _____

Reason: _____

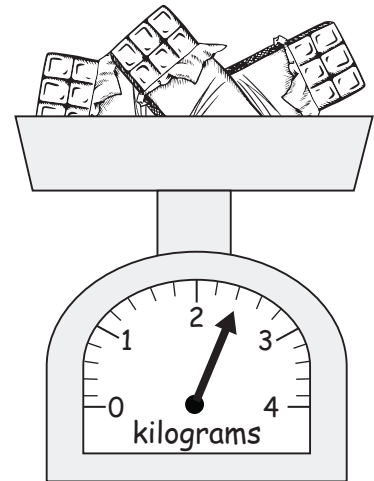
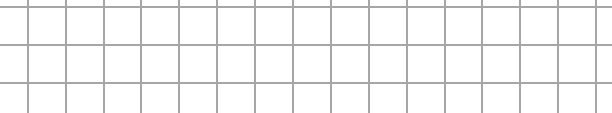


Question 3

(Suggested maximum time: 5 minutes)

- (a)** Three identical bars of chocolate are placed on a scale as shown in the diagram. The arrow on the scale indicates the total weight of the bars.

Find the **weight**, in grams, of one chocolate bar.



- (b)** Sandra and Joe both travel from Cork to Limerick. The distance between the two cities is 100 km.

- (i) Sandra cycles and the journey takes her 5 hours.
Find Sandra's **mean** speed on the journey, in km per hour.

[illegible]

- (ii) Joe drives and the journey takes him 1 hour and 30 minutes. Find Joe's mean **speed** on the journey, in km per hour.

[illegible]

- (iii)** Hence, find the difference between their mean speeds, in km per hour.

[illegible]

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Question 4

(Suggested maximum time: 15 minutes)

During a survey, people were asked to name their favourite social media app. The responses of a random sample of people are shown below.



Snapchat	Facebook	Twitter	Facebook	WhatsApp
Facebook	Facebook	Snapchat	WhatsApp	Twitter
Instagram	WhatsApp	Facebook	Facebook	Snapchat
WhatsApp	Snapchat	Twitter	Snapchat	Facebook
Facebook	Instagram	Snapchat	WhatsApp	Facebook

- (a) Fill in the tally and frequency columns in the table below to show the different responses in the sample.

	Tally	Frequency
Facebook		
Instagram		
Snapchat		
Twitter		
WhatsApp		

- (b)** From the survey, which is the least popular social media app?

Answer =

- (c)** Use your answers to part **(a)** to find the total number of responses in the sample.

[illegible]

- (d)** What is the **mode** of the data?
Give a reason for your answer.

Mode: _____

Reason: _____

- (e) Draw a **bar chart** to display the results of the survey.



- (f) Drawing a bar chart is one way to graphically display this data. Write down another suitable way to graphically display this data.

[illegible]

- (g) Ava says “The data from the survey would give a more accurate representation of the use of social media apps if the survey was only given to students in my school.” Is Ava correct? Give a reason for your answer.

[illegible]

Question 5 (Suggested maximum time: 10 minutes)

The amounts of money eight students spent at the weekend are listed below.

€12	€7	€4	€25	€7	€15	€20	€8
-----	----	----	-----	----	-----	-----	----

- (a) Is this data **categorical** or **numerical**?
Give a reason for your answer.

[illegible]

- (b)** Is this data **discrete** or **continuous**?
Give a reason for your answer.

[illegible]

- (c) Find the **median** amount of money spent.

[illegible]

- (d) Find the **mean** amount of money spent.

[illegible]

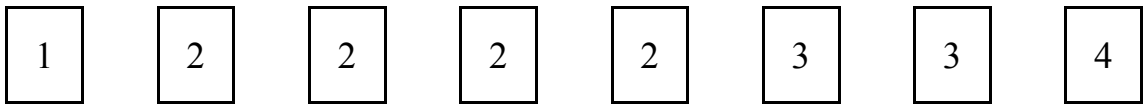
- (e) Tom, another student, also spent some money at the weekend. If the amount he spent was included in the data above, the **range** of amounts of money spent would be €23. Write down the minimum amount of money Tom spent.

[illegible]

Question 6

(Suggested maximum time: 5 minutes)

Sally puts eight cards into a bag. Each card has a number on it, as shown below.

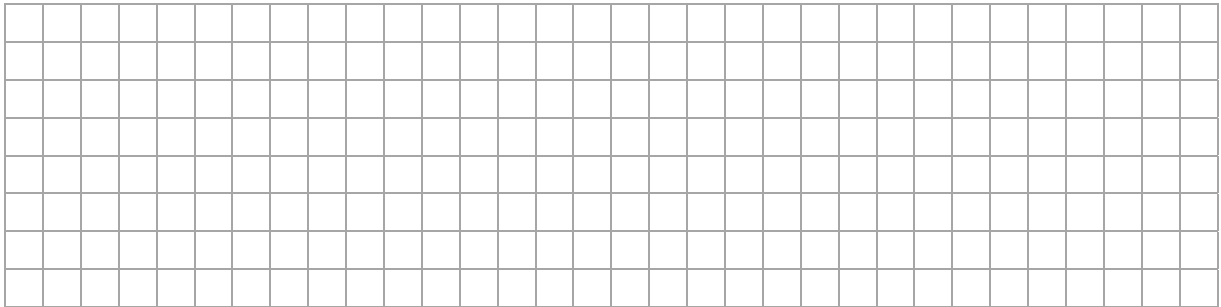
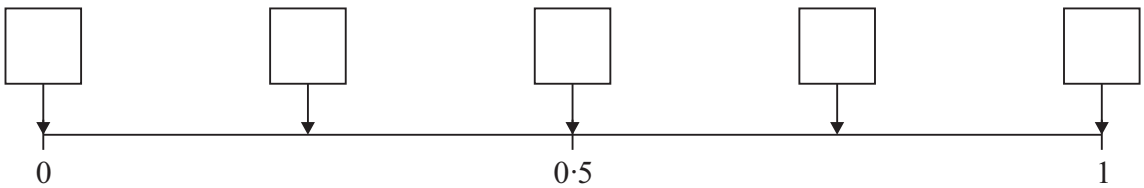


She picks one card at random from the bag.

- (a) Estimate the probability for each of the events **A**, **B**, **C**, **D** and **E** listed below. Write your answers into the table.

Letter	Event	Probability
A	The card has a 3 on it.	
B	The card has a 2 on it.	
C	The card has a 5 on it.	
D	The card does not have a 3 on it.	
E	The card has a number on it.	

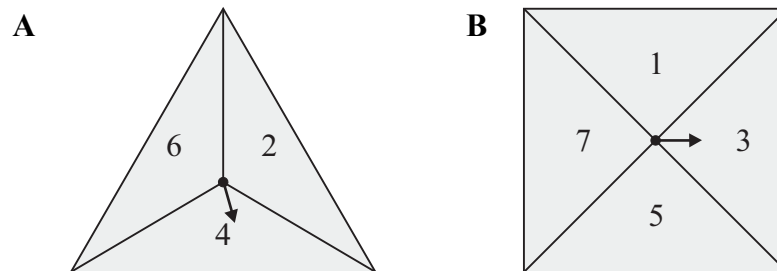
- (b) For each of the events **A**, **B**, **C**, **D** and **E** listed above, place the letter in the most appropriate position on the probability scale below.



Question 7

(Suggested maximum time: 10 minutes)

- (a)** Marie spins two fair spinners as shown.
The results of the two spinners are then added together.
Spinner A has three equal sections labelled 2, 4 and 6.
Spinner B has four equal sections labelled 1, 3, 5 and 7.



- (i) Complete the table below to show all the possible outcomes. Two are already filled in for you.

		Spinner A		
		2	4	6
Spinner B	1	3		
	3			
	5		9	
	7			

- (ii) How many possible outcomes are there?

[illegible]

- (iii) Each outcome is equally likely.
What is the **probability** that the outcome is a 7?

[illegible]

Answer =

- (iv)** What is the **probability** that the outcome is less than 10?

[illegible]

Answer =

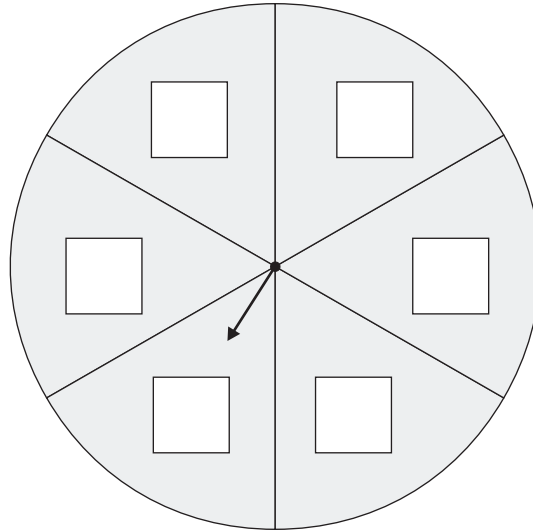
- (b)** Josh makes another spinner. It is also a fair spinner, divided into six equal sectors as shown below.

The spinner needs the letters ***A***, ***B*** and ***C*** to be printed on it.

The probability that the spinner will land on A is $\frac{1}{2}$.

The probability that the spinner will land on **C** is greater than the probability that the spinner will land on **B**.

Complete Josh's spinner by filling the letters into the boxes below.

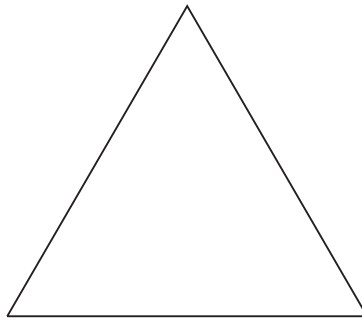
A large grid of 20 columns and 10 rows, intended for drawing. The grid is composed of thin black lines forming a series of small squares.

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Question 8

(Suggested maximum time: 5 minutes)

An equilateral triangle is shown.



- (a)** Explain what it means to say that a triangle is ‘an equilateral triangle’.

[illegible]

- (b)** How many axes of symmetry does an equilateral triangle have? Answer =

- (c)** Draw in all its axes of symmetry in the equilateral triangle above.

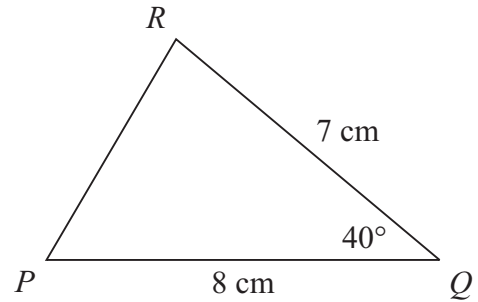
- (d) One side of the equilateral triangle measures 5.8 cm. Find the **perimeter** of the triangle in cm.

[illegible]

Question 9

(Suggested maximum time: 10 minutes)

- (a) Shown is a sketch of the triangle PQR .
 $|PQ| = 8 \text{ cm}$, $\angle RQP = 40^\circ$ and $|QR| = 7 \text{ cm}$.



- (i) **Construct** an accurate drawing of the triangle PQR in the box below.
 Show your construction lines clearly.

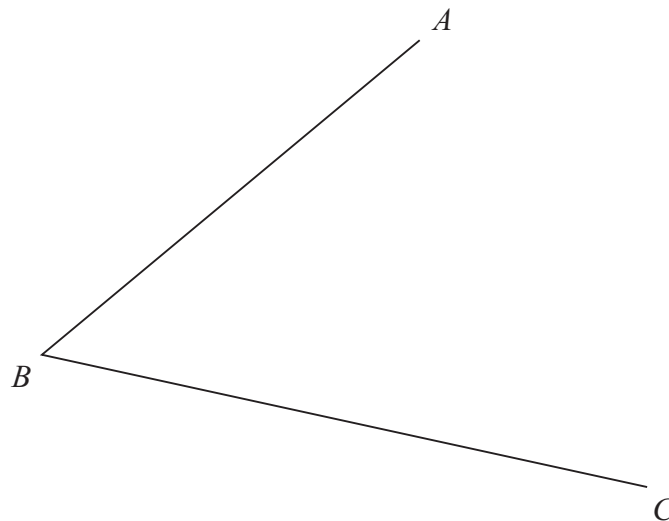
Construction:

- (ii) From your construction, measure $|PR|$.

Answer =

cm

- (b) Using only a **compass** and a **straight-edge**, construct the bisector of the angle ABC .
 Show all construction lines clearly.

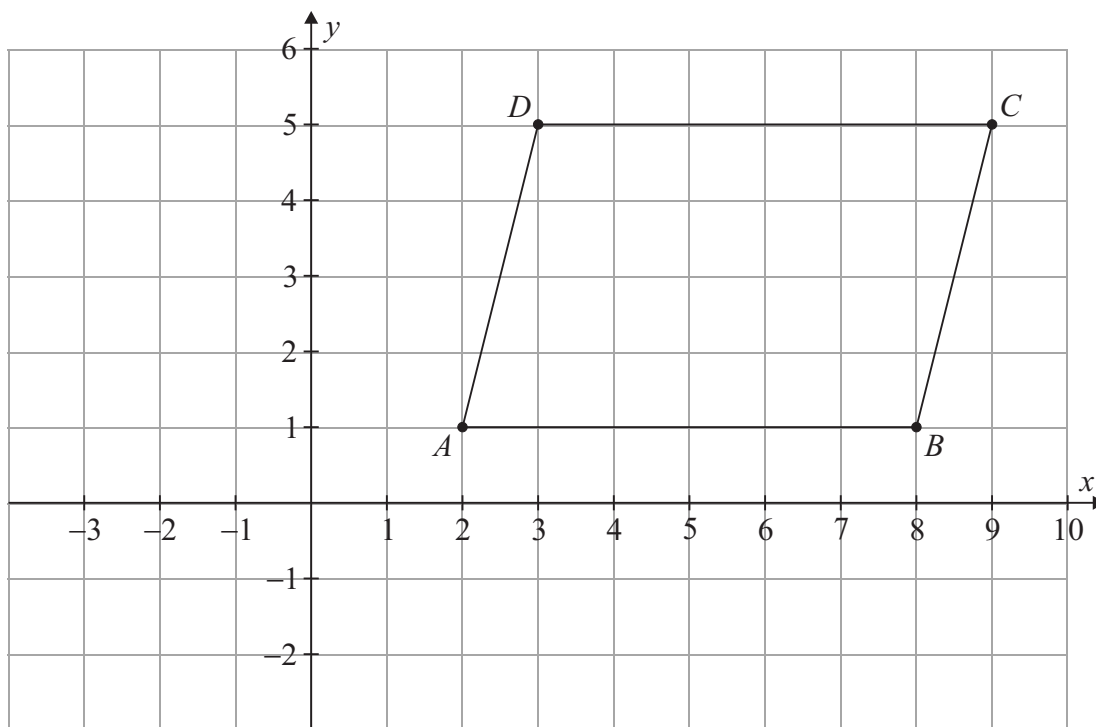


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Question 10

(Suggested maximum time: 10 minutes)

The parallelogram $ABCD$ is shown on the co-ordinate diagram below.



- (a) Write down the co-ordinates of the points A , B , C and D .

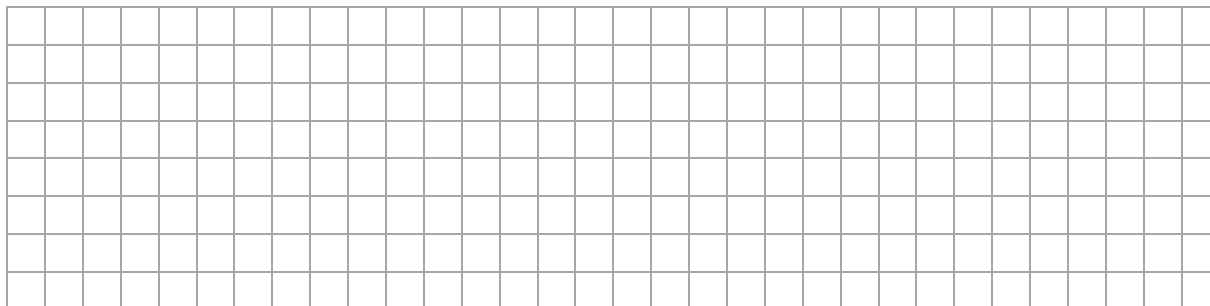
$A = \left(\quad , \quad \right)$ $B = \left(\quad , \quad \right)$
 $C = \left(\quad , \quad \right)$ $D = \left(\quad , \quad \right)$

- (b) On the co-ordinate diagram, **draw** in the diagonals of the parallelogram $ABCD$.

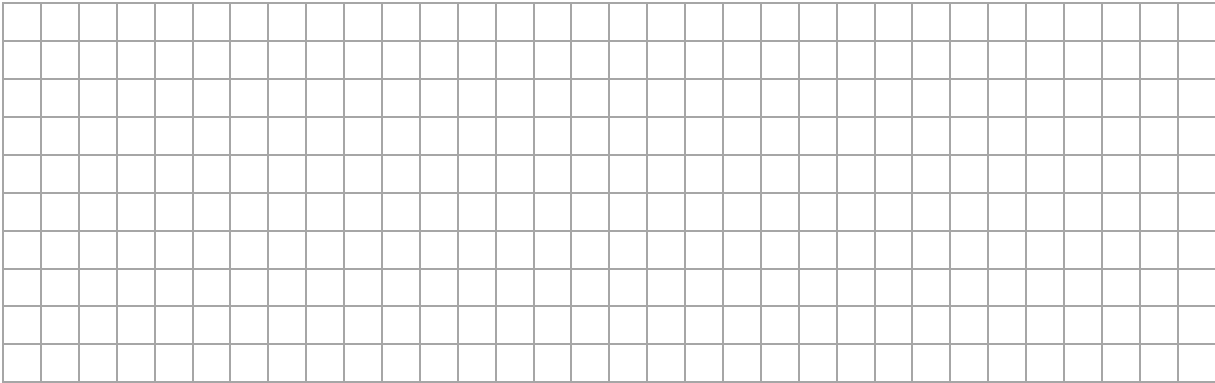
- (c) Hence, write down the co-ordinates of M , the point of intersection of the diagonals.

$M = \left(\quad , \quad \right)$

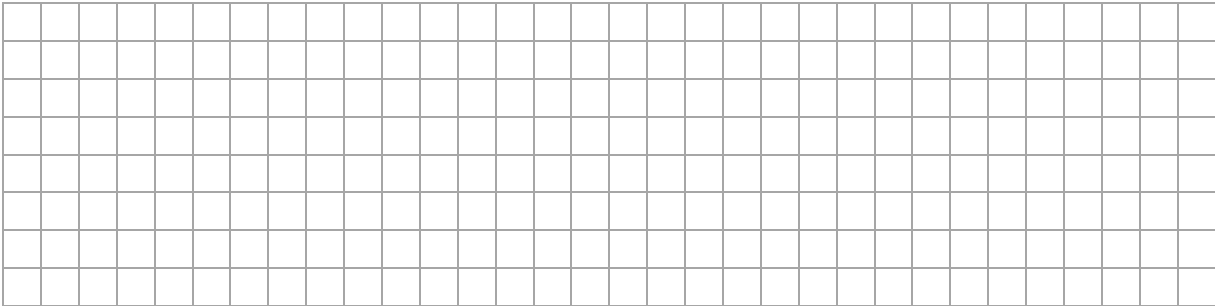
- (d) Find the **slope** of the line AD .



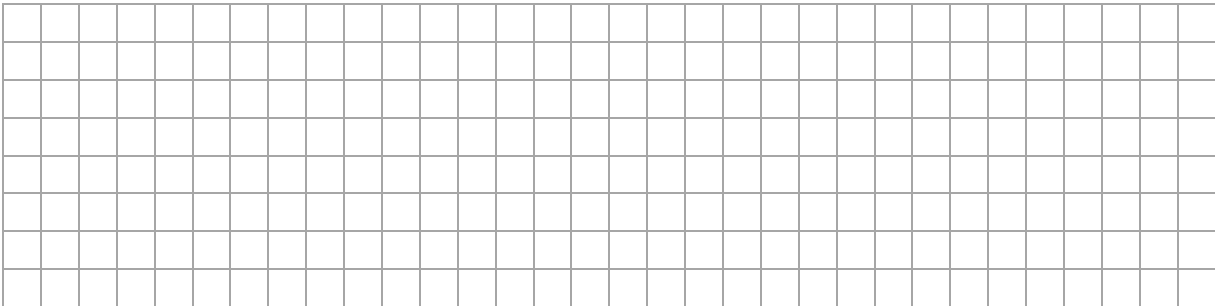
(e) Find the **length** of $[AC]$. Give your answer in the form $\sqrt{x}$, $x \in \mathbb{N}$.



(f) Find the **area** of the parallelogram $ABCD$.



(g) Write down, in your own words, any theorem you know about parallelograms.



Question 11

(Suggested maximum time: 10 minutes)

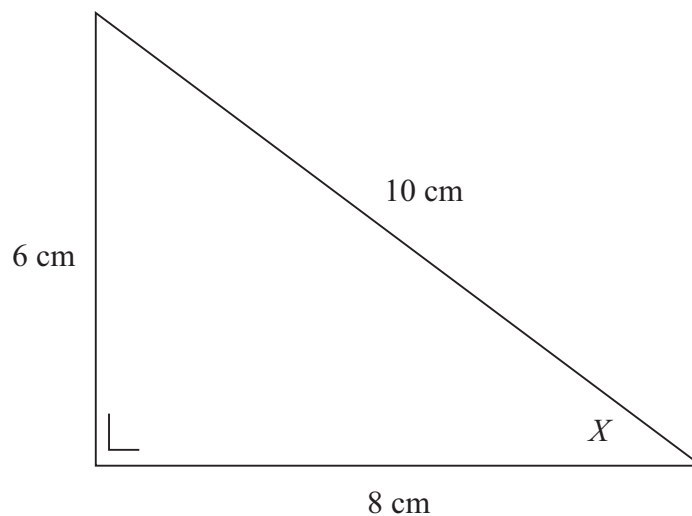
- (a) Use your calculator to find the following trigonometric ratios.
Write each answer correct to three decimal places.

$$\tan 43^\circ = \boxed{}$$

$$\cos 26^\circ = \boxed{}$$

$$\sin 17^\circ = \boxed{}$$

- (b) The diagram below shows the angle X in a right-angled triangle.
The lengths of the sides are also shown.



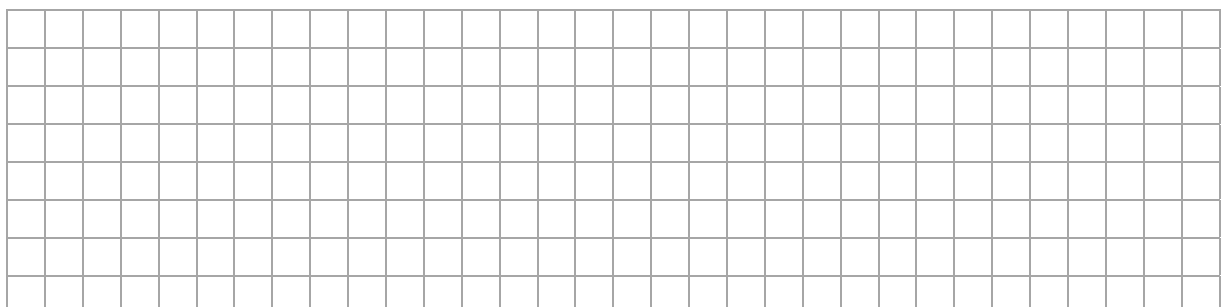
- (i) Write down the length of the side opposite to the angle X , the length of the side adjacent to the angle X and the length of the hypotenuse, in the spaces provided below.

Opposite = cm

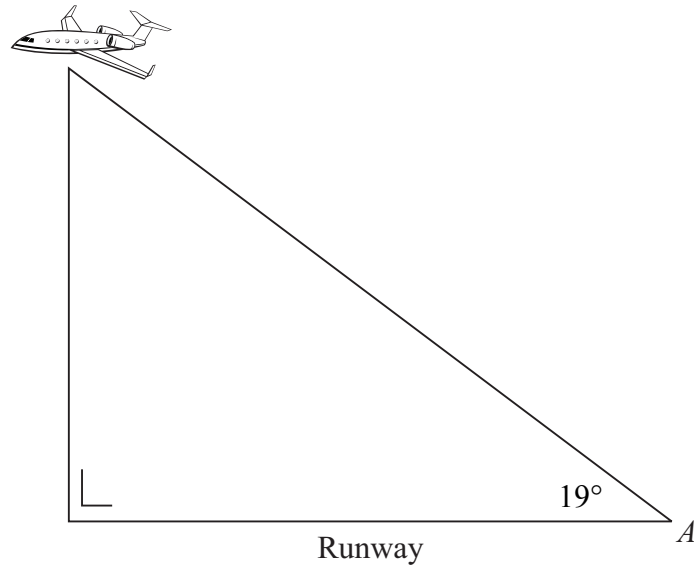
Adjacent = cm

Hypotenuse = cm

- (ii) Use the **Theorem of Pythagoras** to show that the given lengths of the sides of the right-angled triangle are correct.

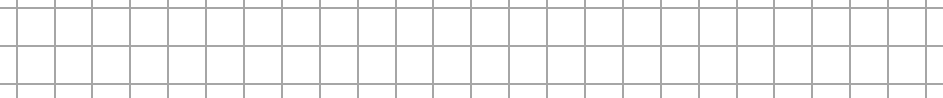


- (c) An aeroplane takes off on a runway from point A at an angle of 19° to the level ground. The mean take-off speed is 75 metres per second.



- (i)** How far does the plane travel in the first 20 seconds?

- (ii) Find the height of the aeroplane above ground level after 20 seconds. Give your answer correct to the nearest metre.

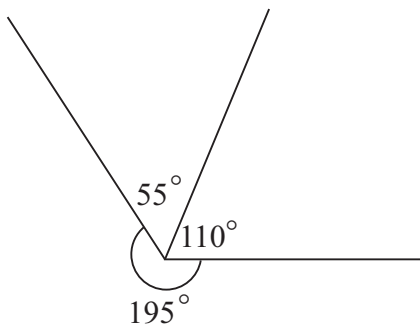


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Question 12

(Suggested maximum time: 10 minutes)

- (a)** Three angles are drawn at a point as shown.



- (i) Write down the measure of the **acute** angle.

Acute angle =

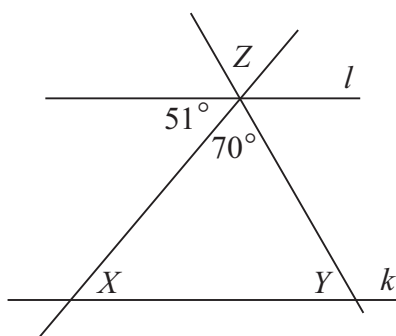
- (ii) Write down the measure of the **reflex** angle.

Reflex angle =

- (iii) Write down the sum of the three angles.

Sum =

- (b)** In the diagram, the line l is parallel to the line k .



- (i) Write down the size of the angle X .

[illegible]
$$|\angle X| =$$


- (ii) Write down the size of the angle Y .

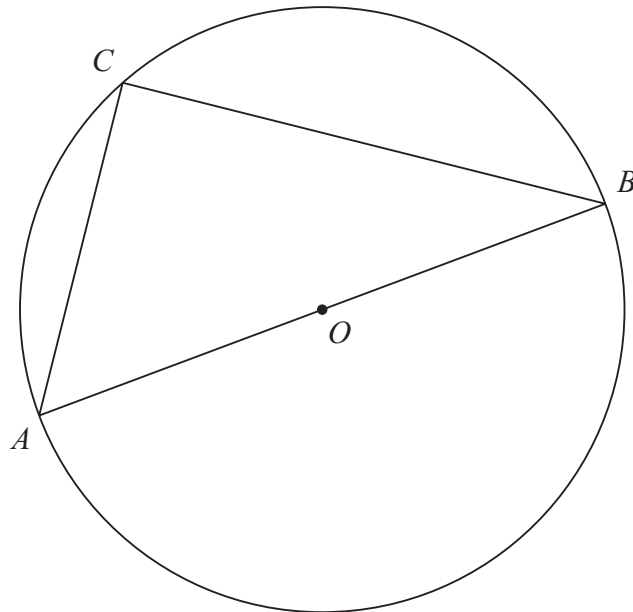
[illegible]
$$|\angle Y| =$$

9

- (iii) Write down the size of the angle Z .

[illegible]
$$|\angle Z| =$$

- (c) A circle with centre O has points A , B and C on the circumference is shown below.



- (i) Write down the measure of the angle ACB .
Give a reason for your answer.

$$|\angle ACB| = \quad \circ$$

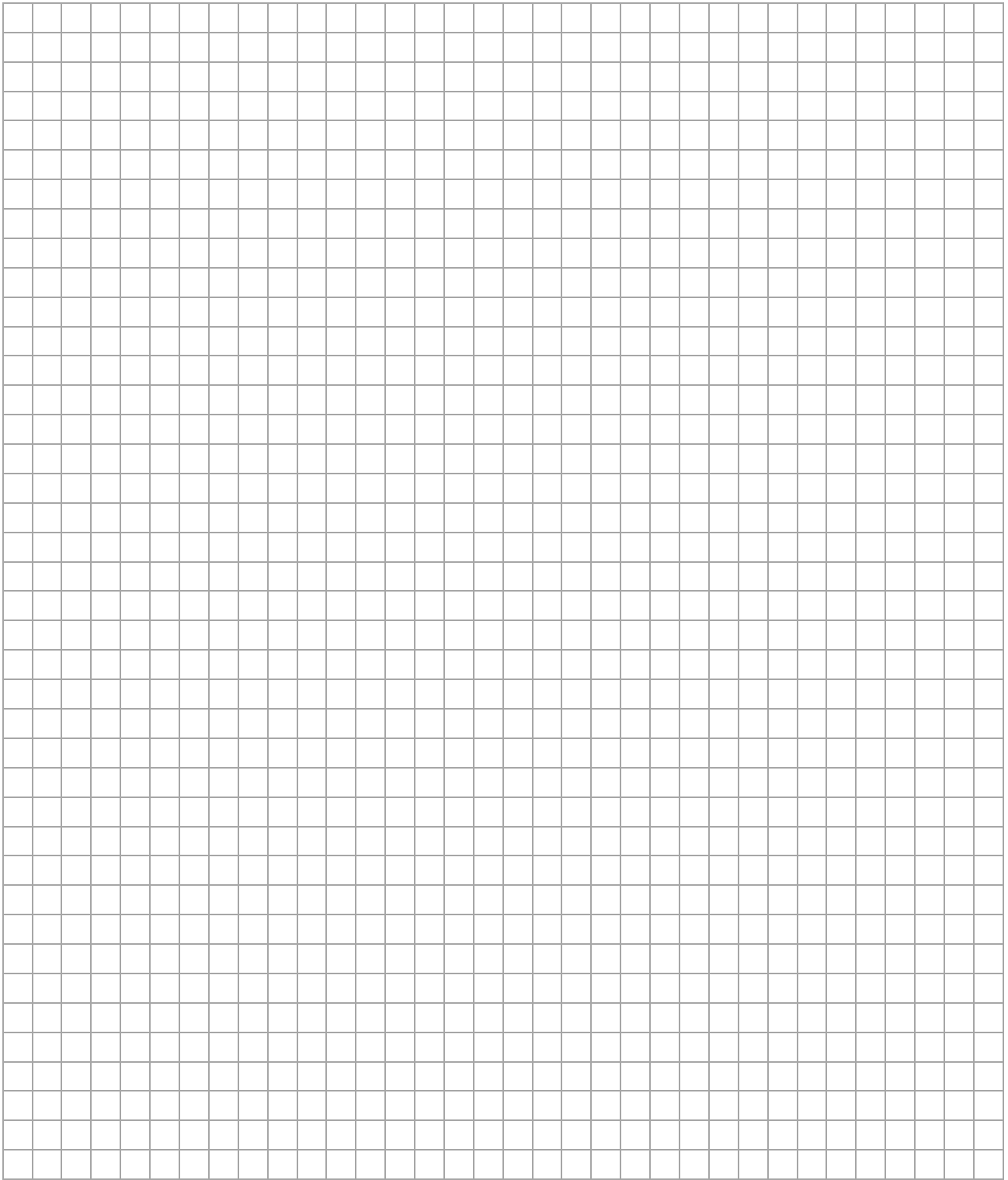
Reason:

- (ii) Using your **ruler**, find the length of the radius of the circle. Give your answer correct to the nearest cm.

Radius = _____ cm

- (iii) Work out the **area** of the circle.
Give your answer in cm^2 , correct to two decimal places.

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Pre-Junior Certificate, 2017 – Ordinary Level

Mathematics – Paper 2

Time: 2 hours

